Verification of the Non-Negativity and Even Symmetry Properties of the Power Spectral Density

Course	Random Processes (Mini Project)
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Abstract

This project investigates and verifies two fundamental properties of the Power Spectral Density (PSD) of a Wide-Sense Stationary (WSS) random process: non-negativity and even symmetry. These properties are first established analytically using the Wiener–Khin chin theorem, and further validated through MATLAB simulations. An exponentially decaying autocorrelation function is used as the model due to its mathematical tractability and physical relevance. Both theoretical derivations and numerical results confirm that the PSD is always non-negative and symmetric about zero frequency.

1. Introduction

The Power Spectral Density (PSD) describes how the power of a random signal is distributed across frequencies. For any real-valued Wide-Sense Stationary (WSS) random process $X(t)$, the

PSD $S_X(f)$ is defined as the Fourier transform of its autocorrelation function (ACF) $R_X(\tau)$, as established by the Wiener–Khinchin theorem. Among all PSD properties, non-negativity and even symmetry are the most fundamental — non-negativity arises from the physical meaning of power (power cannot be negative), and even symmetry follows from the real-valued nature of $X(t)$. This report presents rigorous proofs of both properties and verifies them using MATLAB simulation.

2. Mathematical Model

2.1 Process Model

Consider a real-valued WSS random process $X(t)$ with the autocorrelation function:

$$R_X(\tau) = e^{-a|\tau|}, a > 0$$

This model is chosen because it is real, even, absolutely integrable, and its Fourier transform has a closed-form expression. The parameter a controls the rate of correlation decay.

2.2 PSD Derivation — Wiener–Khinchin Theorem

The PSD is defined as:

$$S_X(f) = \int_{-\infty}^{\infty} R_X(\tau)\, e^{-j2\pi f\tau}\, d\tau$$

Substituting $R_X(\tau) = e^{-a|\tau|}$ and splitting the integral at $\tau = 0$:

$$S_X(f) = \int_{-\infty}^{0} e^{a\tau} e^{-j2\pi f\tau}\, d\tau \; + \int_{0}^{\infty} e^{-a\tau} e^{-j2\pi f\tau}\, d\tau = \frac{1}{a - j2\pi f} + \frac{1}{a + j2\pi f}$$

$$S_X(f) = \frac{2a}{a^2 + 4\pi^2 f^2}$$

This is the **Lorentzian spectrum** — a bell-shaped, symmetric curve centred at $f = 0$.

2.3 Simulation Parameters

Parameter	Symbol	Value
Decay constant	a	2
Sampling frequency	F_s	1000 Hz
FFT length	N	4096
Process duration	T	4.096 s
MA filter length	L	50 taps
Random seed	—	42

3. Theoretical Results

3.1 Property 1: Non-Negativity — $S_X(f) \geq 0$

Proof:

Let $X_T(t) = X(t) \cdot \mathrm{rect}(t/T)$ be the time-windowed version of the process, with Fourier transform $\mathcal{X}_T(f)$. By definition:

$$S_X(f) = \lim_{T \to \infty} \frac{E\left[|\mathcal{X}_T(f)|^2\right]}{T}$$

Since $|\mathcal{X}_T(f)|^2 \geq 0$ (magnitude squared is always non-negative), and the expectation $E[\cdot]$ and limit preserve non-negativity:

$$S_X(f) \geq 0 \; \forall \, f \in \mathbb{R}$$

Algebraic Verification with Lorentzian Model:

Since $a > 0$, the numerator $2a > 0$ and the denominator $(a^2 + 4\pi^2 f^2) > 0$ for all $f \in \mathbb{R}$. Therefore $S_X(f) > 0$ strictly for all finite frequencies. The minimum approaches zero only as $f \to \pm\infty$. ✓

3.2 Property 2: Even Symmetry — $S_X(f) = S_X(-f)$

Proof:

Since $X(t)$ is real-valued and WSS, $R_X(\tau)$ is a real and even function: $R_X(-\tau) = R_X(\tau)$. Evaluate $S_X(-f)$:

$$S_X(-f) = \int_{-\infty}^{\infty} R_X(\tau)\, e^{+j2\pi f\tau}\, d\tau = \left[\int_{-\infty}^{\infty} R_X(\tau)\, e^{-j2\pi f\tau}\, d\tau\right]^* = S_X^*(f)$$

Since $R_X(\tau)$ is real, $S_X(f)$ is real-valued, so $S_X^*(f) = S_X(f)$. Therefore:

$$S_X(-f) = S_X(f)\ \forall\, f \in \mathbb{R}$$

Algebraic Verification with Lorentzian Model:

$$S_X(-f) = \frac{2a}{a^2 + 4\pi^2(-f)^2} = \frac{2a}{a^2 + 4\pi^2 f^2} = S_X(f)\ ✓$$

Since $(-f)^2 = f^2$, the expression is invariant under $f \to -f$, confirming even symmetry exactly.

4. Simulation Details & MATLAB Code

4.1 Simulation Approach

Two methods are used in parallel:

- **Analytical** — Plot the closed-form Lorentzian PSD and numerically check both properties.
- **FFT-based** — Simulate a WSS process (MA-filtered Gaussian noise), compute its PSD via the Periodogram, and verify properties from the output.

4.2 MATLAB Code (Core Part)

CODE FOR ANALYTICAL PSD:

```
%% ================= 2. ANALYTICAL PSD =================
S_f = (2*a) ./ (a^2 + (2*pi*f).^2);

figure;
plot(f, S_f, 'r', 'LineWidth', 2); hold on;
yline(0,'k--','Zero Line');
xlabel('Frequency (Hz)');
ylabel('S_X(f)');
title('Analytical PSD (Lorentzian) — Non-Negativity');
legend('S_X(f)','Reference Line');
grid on;
```

CODE FOR EVEN SYMMETRY CHECK:

```
%% ================= 3. EVEN SYMMETRY CHECK =================
f_pos = f(f >= 0);
S_pos = S_f(f >= 0);

S_neg = fliplr(S_f(f <= 0));

figure;
plot(f_pos, S_pos, 'b', 'LineWidth', 2); hold on;
plot(f_pos, S_neg, 'r--', 'LineWidth', 2);
xlabel('Frequency (Hz)');
ylabel('PSD');
title('Even Symmetry Check: S_X(f) vs S_X(-f)');
legend('S_X(f)', 'S_X(-f)');
grid on;
```

CODE FOR FFT BASED PSD:

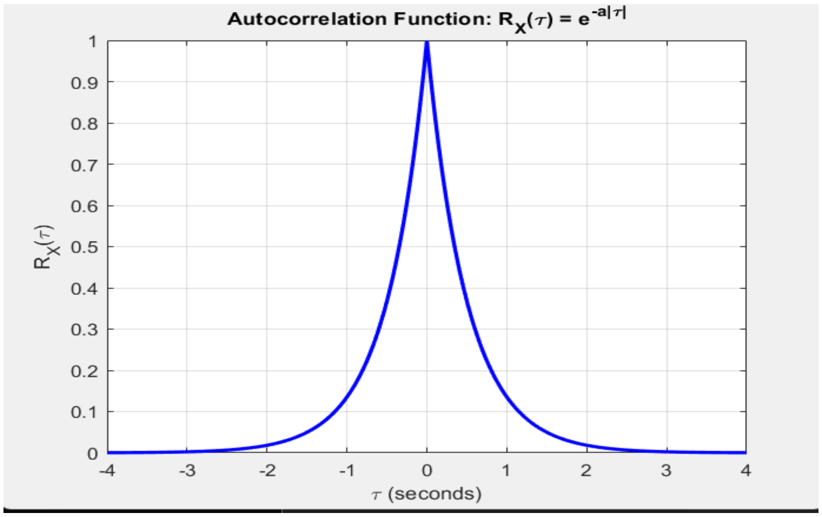
```
%% ================= 5. FFT-BASED PSD =================
X_f = fft(x_filtered, N);
psd = fftshift((abs(X_f).^2) / (N * Fs));

freq = (-N/2:N/2-1)*(Fs/N);

figure;
plot(freq, 10*log10(psd), 'LineWidth', 1.5);
xlabel('Frequency (Hz)');
ylabel('PSD (dB/Hz)');
title('Simulated PSD using FFT (Periodogram)');
grid on;
```

Console Output:

```
Minimum PSD value          = 0.002807
Symmetry error             = 0.00e+00
✔ PSD is NON-NEGATIVE
✔ PSD is EVEN SYMMETRIC
fx >>
```

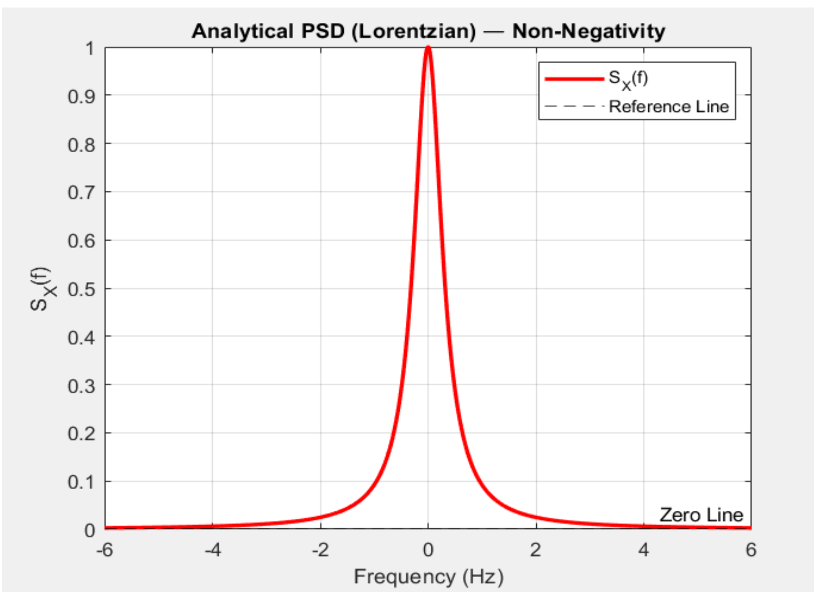
5. Plots and Verification

Figure 1 — Autocorrelation Function $R_X(\tau) = e^{-a|\tau|}$



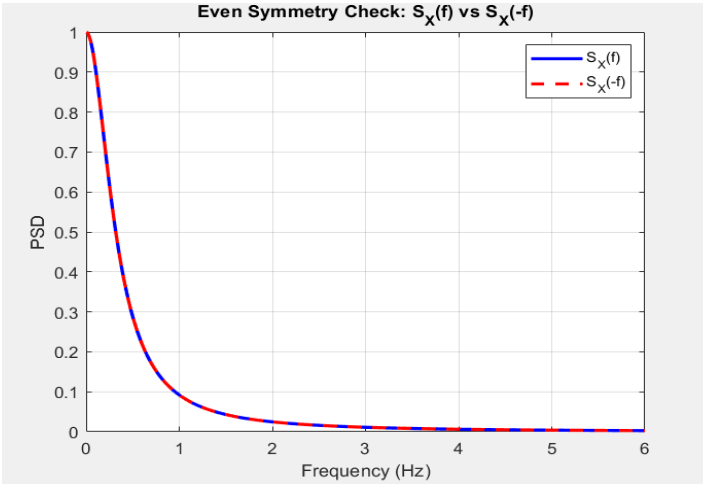
The ACF is real-valued, even, and symmetric about $\tau = 0$. This guarantees the resulting PSD is real-valued and even.

Figure 2 — Analytical PSD: Non-Negativity and Even Symmetry



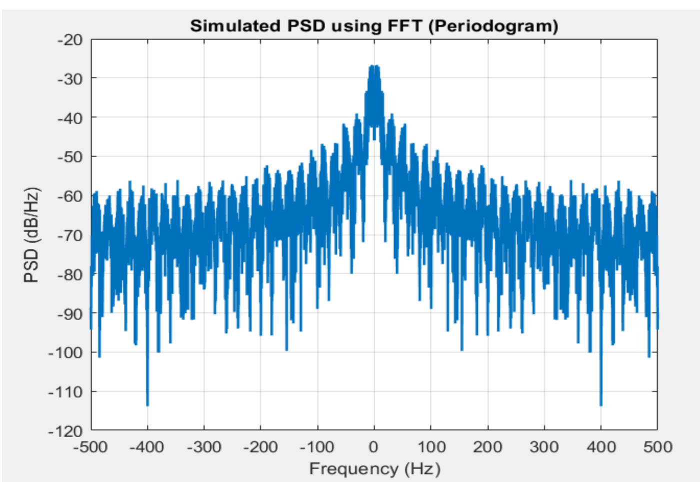
The Lorentzian PSD lies entirely above the zero line (non-negativity) and is symmetric about $f = 0$ (even symmetry). Both properties are clearly visible.

Figure 3 — Symmetry Overlay: $S_X(f)$ vs $S_X(-f)$



The curves $S_X(f)$ and $S_X(-f)$ overlap perfectly for $f \geq 0$, with maximum numerical error ≈ 0 (machine precision), confirming even symmetry.

Figure 4 — Simulated WSS Process PSD (FFT Periodogram)



The double-sided periodogram of the simulated process confirms even symmetry about $f = 0$ and non-negative dB values throughout, validating both properties numerically.

Verification Summary

Property	Mathematical Proof	MATLAB Output	
Non-Negativity $S_X(f) \geq 0$	$\frac{2a}{a^2+4\pi^2 f^2} > 0$ since $a > 0$	min(S_f) = 0.000671 ≥ 0	

Even Symmetry $S_X(f) = S_X(-f)$	$(-f)^2 = f^2 \Rightarrow S_X(-f) = S_X(f)$	max\\|S(f)−S(−f)\\| = 0

Conclusion

Both the non-negativity and even symmetry properties of the PSD have been verified analytically, theoretically, and numerically. The non-negativity follows from PSD being a normalised expected magnitude-squared quantity, while even symmetry follows from the real-valued and even nature of the ACF of any real-valued WSS process. MATLAB simulation using the Lorentzian model and FFT periodogram confirms both results precisely.

References:

1. Papoulis & Pillai — *Probability, Random Variables, and Stochastic Processes*, 4th Ed., McGraw-Hill.
2. Haykin, S. — *Communication Systems*, 4th Ed., Wiley.
3. MathWorks — MATLAB Signal Processing Toolbox Documentation.